AI-Based Intelligent Sales Forecasting & Business Analytics System

1. Introduction

The system is a full-stack AI-based application designed to provide sales forecasting, KPI analytics, and intelligent business insights. It integrates Machine Learning models for numerical forecasting and a Large Language Model (LLM) for generating business explanations.

2. System Overview

The architecture consists of the following core modules:

- Data Module
- Model Module
- Inference & API Module
- Interface Module
- LLM Insight Module

3. Technology Stack

Frontend:

- HTML
- CSS
- JavaScript

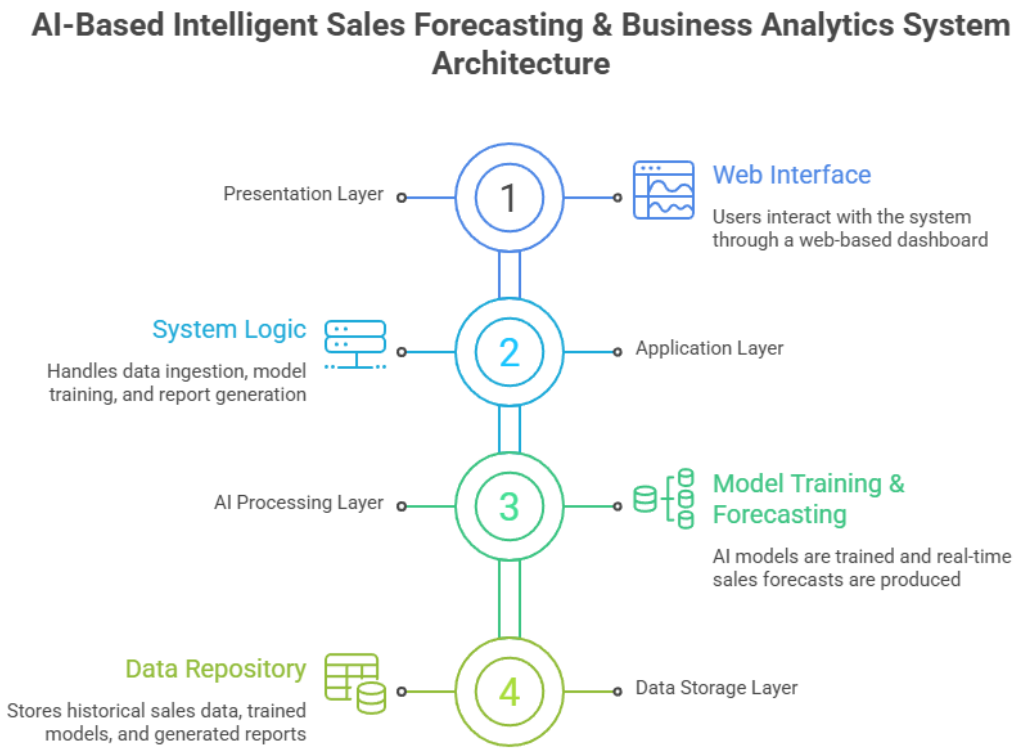
Backend:

- FastAPI (Python)
- Pandas, NumPy
- Scikit-learn
- Statsmodels (ARIMA)
- SQLAlchemy (ORM)
- ReportLab (PDF Generation)
- JWT Authentication

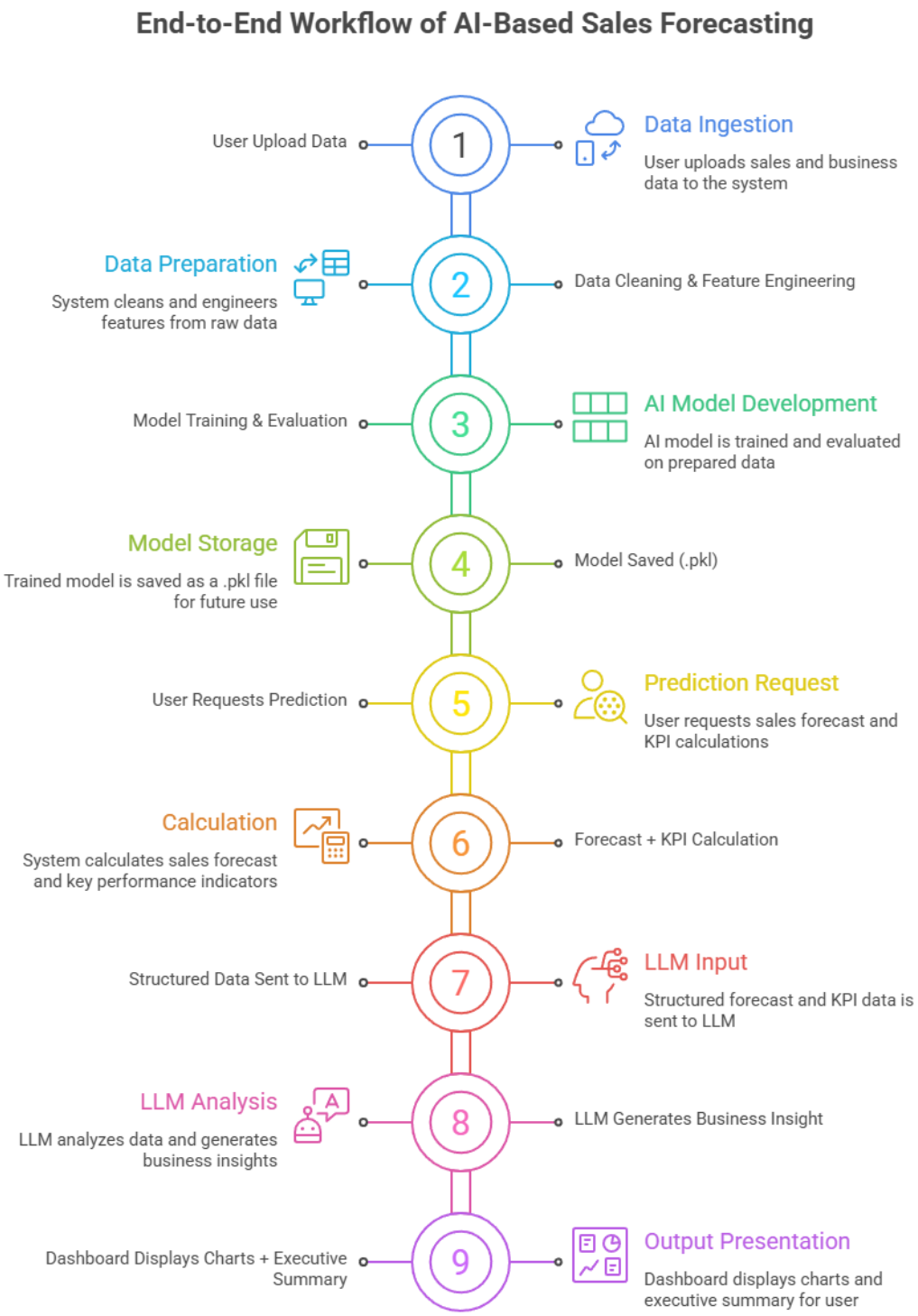
Database:

- PostgreSQL / MySQL

4. System Architecture



5. Workflow


End-to-End Workflow of AI-Based Sales Forecasting
User Upload Data
1
Data Ingestion
User uploads sales and business data to the system
Data Preparation
System cleans and engineers features from raw data
2
Data Cleaning & Feature Engineering
Model Training & Evaluation
3
AI Model Development
AI model is trained and evaluated on prepared data
Model Storage
Trained model is saved as a .pkl file for future use
4
Model Saved (.pkl)
User Requests Prediction
5
Prediction Request
User requests sales forecast and KPI calculations
Calculation
System calculates sales forecast and key performance indicators
6
Forecast + KPI Calculation
Structured Data Sent to LLM
7
LLM Input
Structured forecast and KPI data is sent to LLM
LLM Analysis
LLM analyzes data and generates business insights
8
LLM Generates Business Insight
Dashboard Displays Charts + Executive Summary
9
Output Presentation
Dashboard displays charts and executive summary for user

6. LLM Insight Module

The LLM module enhances the system by generating natural language explanations from structured sales and KPI data. It converts numerical outputs into professional business summaries and answers user queries in plain English.

- Generates executive summaries from KPI data
- Explains forecast trends
- Answers basic business questions from users
- Produces output on the dashboard

7. Conclusion

This system combines traditional machine learning forecasting with LLM-based business intelligence to provide both accurate predictions and meaningful insights. It serves as a modern decision-support platform suitable for industry-level projects.